Summary

- The state is everything that should be preserved when shutting down a service and spinning up a new instance.
- Data services are the stateful components of a cloud native architecture, requiring storage technologies to persist the state.
- Using data services in the cloud is challenging because it’s a dynamic environment.
- Some issues to consider when choosing a data service are scalability, resilience, performance, and compliance with specific regulations and laws.
- You can use data services that are offered and managed by your cloud provider or manage your own, either relying on virtual machines or containers.
- Spring Data provides common abstractions and patterns for accessing data, making it straightforward to navigate the different modules dedicated to relational and non-relational databases.
- The main elements in Spring Data are database drivers, entities, and repositories.
- Spring Data JDBC is a framework that supports integrating Spring applications with relational databases relying on a JDBC driver.
- Entities represent domain objects and can be managed by Spring Data JDBC as immutable objects. They must have the field hosting the primary key annotated with `@Id`.
- Spring Data lets you capture audit metadata whenever an entity is created or updated. You can enable this feature with `@EnableJdbcAuditing`.